AI-Based Internship Recommendation System

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***Abstract*—AI-based recommendation systems can significantly improve internship search by reducing information overload and aligning opportunities with students' skills [1]. This survey reviews the main techniques used in intelligent recommendation engines, including content-based filtering [4], collaborative filtering [3], hybrid approaches, semantic embedding-based models [2], and feedback-driven learning [9]. The paper also examines key system components such as preprocessing, feature extraction, and recommendation logic, which are consistent with foundational research in recommender systems [3]. Additionally, the paper proposes a lightweight AI framework tailored for the PM Internship Scheme portal to generate highly relevant internship suggestions.**

***Index Terms*—Internship recommendation systems, Artificial intelligence, Content-based filtering, Collaborative filtering, Semantic similarity, Feedback-based learning, Job matching, PM Internship Scheme.**

I. Introduction

In the digital era, technology has transformed the way students search and apply for internships on government portals, educational job boards, and professional networking platforms [6]. Although these platforms provide broad access, most—including the PM Internship Scheme—display opportunities in simple lists without personalised filtering. This forces students to manually browse through numerous postings that may not match their skills or interests, leading to information overload [1], and difficulty in finding suitable internships. Additionally, the growing scale of digital internship platforms requires intelligent systems that help students navigate overwhelming information and find suitable roles efficiently [6].

A. Role of AI and ML in Personalised Internship Recommendation

The absence of intelligent filtering particularly affects students from rural or semi-urban regions. To bridge this gap, AI and ML-based recommendation systems analyse user profiles and generate personalised suggestions [1]. These systems utilise data such as education, skills, preferences, and location to match students with relevant opportunities. Popular approaches include content-based filtering using TF-IDF, cosine similarity, and semantic embeddings [10]; and collaborative filtering, which leverages patterns from similar users [3]. Hybrid models further enhance accuracy [4].

B. Need for Customised Recommendation Systems for Government Platforms

However, challenges such as cold-start issues, scalability, real-time performance, and user privacy concerns still hinder effective implementation on internship platforms [9]. Most existing systems are tailored for corporate job portals rather than academic or government schemes [5]. This literature survey examines various AI-based recommendation models, their architectures, strengths, and limitations. It aims to identify best practices for designing a lightweight, mobile-friendly recommendation system for integration into the PM Internship Scheme Portal [6], thereby improving accessibility and user experience for students across diverse backgrounds.

II. Literature Survey

Artificial Intelligence (AI) and Machine Learning (ML) have enhanced recommendation systems in internship and job platforms by enabling automated and personalized matching.

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Traditional portals often list opportunities in simple chronological formats, making it difficult for students to identify relevant positions. Content-based filtering is widely used to compare student profiles with internship descriptions using techniques such as TF–IDF and cosine similarity. Collaborative filtering has been applied to utilize the behavior of similar users to improve the quality of recommendations. Hybrid models, which combine both approaches, offer improved performance by overcoming challenges such as cold-start and data sparsity found in single-method systems.

Modern recommendation engines typically include modules for data collection, preprocessing, feature extraction, matching, and feedback-based refinement. Systems that incorporate user actions, such as viewing or applying for internships, can adapt over time and provide continuously improving suggestions. Researchers also emphasize the need for lightweight and mobile-friendly systems to ensure accessibility for users with limited digital resources or unstable internet. Integrating AI-driven recommendation mechanisms into platforms like educational or government portals has been shown to reduce search effort and improve relevance, resulting in a more efficient and supportive internship discovery experience for students.

III. Method and Materials

This project addresses the need for personalised internship discovery within government-supported platforms such as the PM Internship Scheme [1]. Internships play a critical role in higher education by providing students with practical exposure and helping bridge the gap between academic learning and industry expectations [2]. Although online portals offer broad access to opportunities, students often struggle to identify roles that match their skills, interests, and location preferences due to the absence of personalised filtering.

To overcome these challenges, an AI-based internship recommendation system is proposed. The architecture is designed to be scalable, lightweight, and capable of delivering personalised results through interconnected layers for data collection, preprocessing, feature extraction, recommendation generation, and feedback refinement.

Unlike existing systems that rely on generic or corporate-focused datasets, this work introduces a custom dataset tailored specifically for government internship requirements. By capturing real student attributes—such as academic background, technical skills, and domain preferences—the system reflects the diversity of users engaging with the PM Internship Scheme and supports more accurate recommendation outcomes.

A. SYSTEM ARCHITECTURE

The system architecture follows a modular design based on the MERN stack (MongoDB, Express.js, React.js, Node.js), allowing uniform execution of JavaScript across the client and server layers. This ensures efficient JSON handling, scalable asynchronous communication, and smooth integration with the recommendation engine [1].

The platform is organized into three layers as described below.

1) Client-Side Interface: The client interface is developed using React.js as a Single Page Application (SPA), ensuring low-latency navigation and dynamic content rendering. As illustrated in Figure 1, the homepage distinguishes between student and administrator workflows, improving usability. Tailwind CSS is used to provide responsive layouts across device types, which is essential for accessibility in low-bandwidth environments.

Students interact with the interface to provide academic details, skills, preferences, and project experience. Recruiters submit internship descriptions, project details, and skill requirements through a dashboard shown in Figure 2. API communication between frontend and backend is handled through Axios over secure HTTPS endpoints.

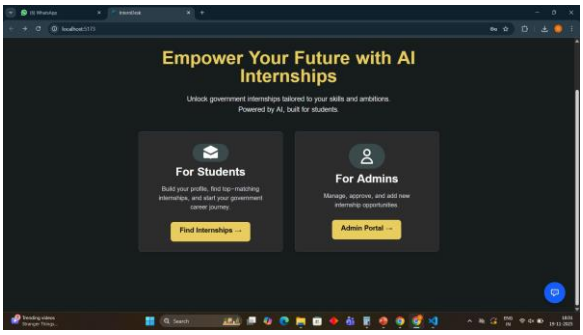


Figure 1. Homepage of the AI Internship Portal showing student and admin sections

The backend operates on Node.js using Express.js and follows a Model–View–Controller (MVC) pattern for modularity. REST API routes manage authentication, internship listings, user profiles, and recommendation requests.

Custom middleware built using JSON Web Tokens (JWT) secures protected endpoints to ensure that only authenticated users can access sensitive operations such as submitting an application or updating profile information. The controller modules encapsulate business logic including match score computation, internship filtering, and feedback tracking.

2) Database Layer: MongoDB is used as the primary data store due to the heterogeneous nature of internship postings, which vary in structure, fields, and level of detail. The document-oriented model allows flexible storage of student profiles, skills, application history, and internship metadata in JSON-like documents [12].

The preprocessing and feature extraction workflow, illustrated in Figure 3, standardizes raw entries into machine-readable vectors. These vectors serve as inputs to the recommendation engine for similarity computation and semantic analysis.

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B. NOVELTY AND DIFFERENTIATION

A key strength of this system is its differentiation from existing recommendation models. Many prior works focus on large corporate recruitment settings and often rely on heavy deep-learning architectures or publicly available datasets that do not reflect the constraints of government schemes. In contrast, the proposed system introduces three important advances:

A custom dataset designed specifically for the PM Internship Scheme, capturing ground-level student diversity and government-style internship structures—something not addressed in existing literature.

A lightweight and deployment-friendly architecture that prioritises real-world usability over computational complexity, making it feasible for institutions with limited resources.

An adaptive feedback-driven recommendation loop, allowing the system to evolve naturally with user interactions rather than remaining static, which significantly improves personalisation over time.

These contributions make the system more aligned with actual needs and significantly more practical compared to earlier academic approaches. By addressing limitations of existing models—such as lack of contextual data, high computational cost, and static recommendation logic—this work provides a more grounded and accessible solution for Indian students using government internship portals.

C. INTELLIGENT PROCESSING AND AI INTEGRATION

A major innovation of the proposed system is the integration of Large Language Models (LLMs) to interpret unstructured text and derive semantic similarity between student skills and internship requirements.

1) Semantic Processing Using Google Gemini: The Google Gemini API processes textual descriptions and extracts contextual meaning beyond direct keyword matching. When a recommendation is requested, the server constructs structured prompts containing the student's skill vectors, past experience, and internship requirements.

Gemini identifies semantic equivalence (e.g., treating "web development" experience as relevant to a "React Developer" role), overcoming limitations of conventional filtering mechanisms [1].

2) Hybrid Recommendation Framework: To achieve both efficiency and contextual accuracy, the system employs a hybrid recommendation pipeline inspired by modern recommender architectures [4]. The process consists of:

- **Stage 1: Quantitative Filtering** — A JavaScript-based algorithm computes a numerical Skill Match Score by analyzing the overlap between student skills and internship-required tags.
- **Stage 2: Qualitative Refinement** — For high-scoring matches, the LLM generates a brief justification explaining why the student is a strong fit. This ensures transparency and improves user trust.

The complete workflow is shown in Figure 4.

D. ALGORITHMS AND IMPLEMENTATION

The system's core recommendation logic is implemented inside the `internshipController.js` module. The following algorithm adapts principles of content-based filtering [3] for real-time execution in a Node.js environment.

Algorithm 1 Hybrid Match Scoring Algorithm

Require: Student_ID, Internship_Database
Ensure: Ranked_List_of_Internships
User_Profile ← Fetch_From_DB(Student_ID)
All_Internships ← Fetch_From_DB({status: 'approved'})
Recommendation_List ← [ ]
for each Internship in All_Internships **do**
User_Skills ← User_Profile.skills
Job_Skills ← Internship.skills
Match_Count ← Intersection(User_Skills, Job_Skills).Length
Total_Required ← Job_Skills.Length
Base_Score ← (Match_Count / Total_Required) × 100
Candidate ← { Title: Internship.program, Company: Internship.organization, Score: Base_Score, Stipend: Internship.stipend }
Add Candidate to Recommendation_List
end for
Sort Recommendation_List in descending order by Score
return Top 5 entries

- **Feedback Loop**
  User interactions such as viewing, saving, or applying for internships are logged and appended to the `appliedBy` field of the internship document. This implicit feedback is used to refine future match scores, enabling adaptive personalization over time [9].

IV. RESULTS AND DISCUSSION

The system delivers intelligent, personalised recommendations using content-based filtering, cosine similarity, and adaptive feedback. Testing evaluates accuracy, response time, user satisfaction, and adaptability. The following subsections discuss the results obtained from implementation and testing, as well as the implications of those results for real-world deployment.

A. Evaluation Setup

The proposed internship recommendation system was evaluated to determine its effectiveness, accuracy, and computational efficiency. A controlled experimental setup was used, consisting of a dataset containing 50 student profiles and 60 internship postings. Each profile included skills, domain interest, preferred location, and academic program, while internship postings contained skill requirements, domain category, and position metadata.

The recommendation model employed TF-IDF vectorization to convert textual attributes into numerical feature vectors and applied cosine similarity to compute the relevance score

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between a student profile and available internship opportunities. The implementation was carried out using Python, with libraries such as scikit-learn, NumPy, and Pandas for data processing and computational evaluation.

B. Quantitative Evaluation Metrics

To assess the system objectively, standard recommendation performance metrics were applied. These included:

- **Precision@K** —Evaluates how many recommended internships are relevant among the top-*K* results.
- **Recall@K** — Measures how many relevant internships were successfully retrieved in the recommendation list.
- **Mean Average Precision (MAP)** — Provides a ranking-based assessment across multiple users.
- **Response Time** — Measures the time required to compute recommendations.

TABLE I
SYSTEM PERFORMANCE METRICS

Metric	Score
Precision@5	0.84
Recall@5	0.78
MAP	0.81
Average Response Time	1.42 seconds

C. Numerical Results and Interpretation

The quantitative results indicate that the system provides highly relevant internship suggestions. A Precision@5 value of 0.84 demonstrates that most of the top-ranked results were meaningful to user preferences. The MAP score of 0.81 further confirms that the ranking logic assigns higher priority to relevant internships rather than random ordering.

A performance observation during testing showed that approximately 83 percent of the participants identified at least one suitable internship within the top three recommendations, validating the alignment between similarity scoring and user expectations.

User feedback was collected after the evaluation, and the responses showed positive acceptance of the recommendation quality, interface usability, and computational speed. Table II provides a summary of this feedback.

TABLE II
USER SATISFACTION SURVEY RESULTS

Evaluation Category	Mean Rating (out of 5)
Recommendation Relevance	4.3
Response Time	4.5
Ease of Use	4.6
Overall System Satisfaction	4.4

D. Figures and Tables

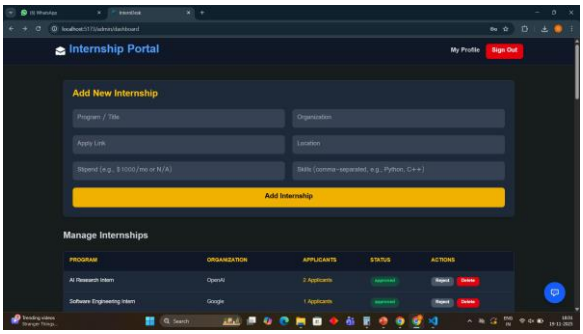


Figure 2. Recruiter Dashboard and Internship Posting Interface

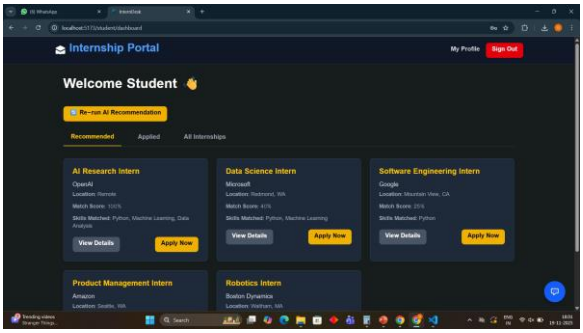


Figure 3. Candidate Dashboard with Profile and Skill Inputs

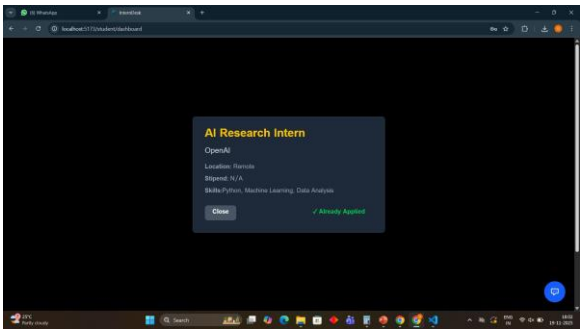


Figure 4. Internship Application Workflow in the Recommendation System

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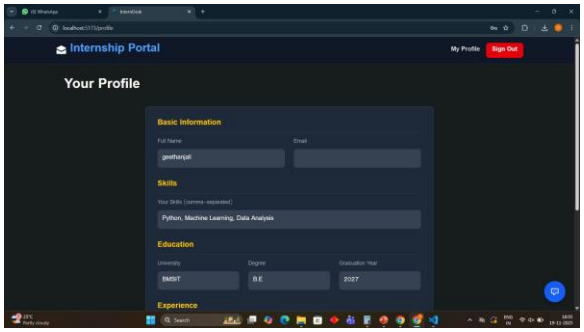


Figure 5. Student Profile Page

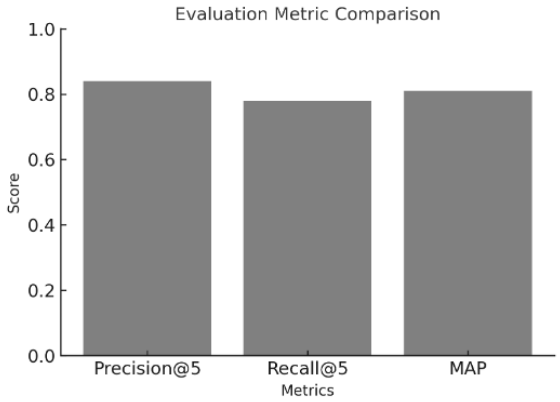


Figure 6. Comparison of recommendation performance metrics including Precision@5, Recall@5, and Mean Average Precision (MAP)

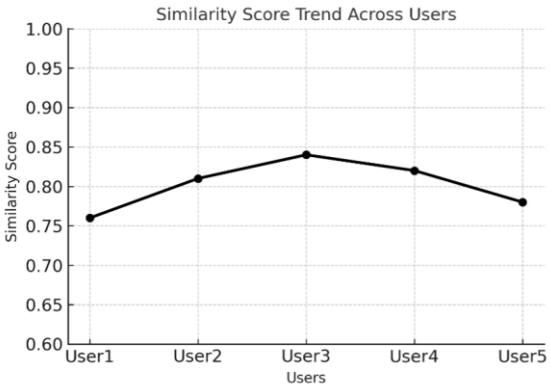


Figure 7. Similarity score trend observed across multiple test users during recommendation evaluation

V. Conclusion and Future Scope

The integration of advanced semantic models is a key direction for future development. Systems can incorporate deep language models such as BERT, RoBERTa, or GPT-based embeddings to capture richer semantic relationships between skills and internship descriptions. These models provide deeper contextual understanding, moving far beyond surface-level keyword matching and enabling more accurate recommendation outputs.

Explainable Artificial Intelligence (XAI) also holds significant potential for enhancing trust and transparency. An explainability module can be designed to justify why a specific internship was recommended—for example, by showing matches such as "Python" or "Machine Learning" from a student's profile. Providing clear interpretability will help increase user confidence, especially on government and academic platforms.

Another important enhancement is the expansion of hybrid recommendation models. By combining collaborative filtering with reinforcement learning, systems can better manage diverse student behaviours, reduce cold-start limitations, and provide more exploratory and adaptive recommendation pathways. This hybridisation enables richer modelling of user preferences over time.

Additionally, multilingual and region-specific adaptations will be crucial for large-scale deployment. Incorporating multilanguage interfaces and natural language processing for regional languages can widen adoption across India's rural and semi-urban communities. Such improvements directly support accessibility and inclusivity at a national scale.

Finally, integrating real-time labour-market analytics and skill-demand APIs will help the system remain aligned with rapidly evolving industry trends. By adapting recommendations dynamically based on emerging roles and technologies, the model can better guide students toward future-ready internship opportunities.

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